Nikhil Mandayam Adyapak

+1(608)395-5104 | nikhiladyapak31@gmail.com | linkedin/in/nikhil-adyapak | nikhiladyapak.github.io | Google Scholar

Summary

Master's student in Data Science at UW-Madison with 2+ years of experience as a Senior Software Engineer building scalable MLOps platforms and distributed AI systems. Specialized in the engineering side of AI, including architecting cloud-native training pipelines on Azure Kubernetes Service (AKS) and optimizing GPU infrastructure. Proven track record of reducing training cycles and engineering efficient data retrieval systems for ADAS.

Actively seeking Fall Co-op (Aug–Dec) opportunities and 2027 Full-Time Roles in Machine Learning Engineering, MLOps, and Data Science.

Education

University of Wisconsin-Madison — Sep 2025 - May 2027
Masters in Data Science — GPA - 3.66/4.00
Coursework: Machine Learning, Foundation Models, Database Design & Implementation, Statistical Models, Statistical Methods, Statistical Inference for Data Science

PES University - Bengaluru, India — Aug 2019 – May 2023
Bachelor of Technology in Computer Science & Engineering — GPA - 3.78/4.00
Specialization: Machine Intelligence & Data Science
Coursework: Algorithms, Data Structures, Data Science, Linear Algebra, Machine Learning, Big Data, Cloud Computing

Technical Skills

Programming languages: Python, C, C++, Java, SQL, R
ML & Statistics: PyTorch, TensorFlow, HuggingFace, Transformers, LLM, CNN, GAN, scikit-learn, OpenCV, matplotlib
MLOps & Workflows: Ray, MLFlow, DVC, Argo Workflows, Azure ML Studio, Prometheus, Grafana
Cloud: Azure, AWS, AKS, Docker, Terraform, Azure DevOps, GitHub Actions, Linux
Databases & Tools: ElasticSearch, PostgreSQL, MySQL, MongoDB, FastAPI, Flask, Streamlit, GitHub

Research Experience

Bosch Global Software Technologies (BGSW) — Bengaluru, India
Senior Software Engineer – AI Systems for Advance Driver Assisted Systems (ADAS) — Aug 2023 – Aug 2025

- Engineered **scene-understanding** image retrieval pipeline using foundation models (**Mask2Former, Depth-Anything, OWL-ViT, CLIP**) to generate scene graphs, curating fine-grained datasets for ADAS corner cases.
- Benchmarked **Occlusion Estimation** models on the **nuScenes** dataset by engineering custom LiDAR-depth evaluation pipelines to generate metrics and visualizations.
- Enhanced ADAS image dataset resolution by **4x** utilizing **ESRGAN** transfer learning on curated datasets.
- Researched optimal deployment strategies for multi-tenant **Ray Clusters** on **on-premise workstations** and **Azure Kubernetes Service (AKS)** to efficiently support **ML Workloads**.
- Integrated **CLIP embedding** re-use with **80%** cross-version alignment, avoiding full re-indexing of legacy image vectors.

Center for Information Security Forensics and Cyber Resilience, PES University — Bengaluru, India
Summer Research Intern; Advisor: Prof. Vineetha B, Dr. Prasad B Honnavalli — May 2021 – Nov 2021

- Developed a novel cryptanalysis algorithm in **Python** to decipher single columnar transposition ciphers without prior knowledge of the encryption key or its length.
- Engineered an optimization approach reducing computational overhead by dynamically determining key lengths via matrix permutations and longest-prefix dictionary matching.

Industry Experience

Ciroos — Pleasanton, CA (SF Bay Area)
AI Intern — May 2026 – Aug 2026

- Conducted performance evaluations (**evals**) of AI-driven Site Reliability Engineering (SRE) agents by simulating infrastructure faults across Kubernetes and multi-cloud environments (**AWS, GCP, Azure**).
- Authored and fine-tuned detailed instruction pipelines that guide AI agents to accurately perform Root Cause Analysis (**RCA**) and fault diagnosis across varied system domains.

Bosch Global Software Technologies (BGSW) — Bengaluru, India
Senior Software Engineer – AI Systems for Advance Driver Assisted Systems (ADAS) — Aug 2023 – Aug 2025

- Delivered **MLOps platform** for pedestrian detection (**Azure ML Studio + MLFlow**) with end-to-end **ML lifecycle**; enabled **multi-GPU training** (**3x** faster) and cut GPU costs by **60%**.
- Designed and scaled a **Ray Cluster** based distributed **ML training** system on **Azure Kubernetes Service (AKS)**, reducing retraining cycles from **4 weeks** to **1 week** (**75% faster**) and supporting **30+** teams.
- Architected a containerized **FastAPI** backend for a **60M+ vector image-search engine**, deploying **CLIP embeddings** and **ElasticSearch** to reduce edge-case triage by **95%** (4 hours to 10 mins).

- Automated **ML pipeline** CI/CD workflows on **AKS** (**Terraform, Helm, Argo, GitHub Actions**); integrated **Prometheus** alerting, accelerating experiment setup for **multi-tenant** ADAS teams from **3 days** to **5 minutes**.
- Deployed on-prem **Ollama LLM** with Python FastAPI wrapper, enabling private, team-wide LLM access (**150+ users**), zero external data egress.

Machine Learning Operations (MLOps) Intern Jan 2023 – May 2023

- Built on-premise **MLOps** pipelines (**DVC**, **MLflow**) and accelerated **multi-GPU** object detection training (YOLOv5, Detectron2) on **Azure ML Studio**, cutting setup time from **1 week** to **1 day** (**85% faster**) and training time by **67%**.
- Containerized models (**Docker**) and engineered CI/CD workflows via **Azure DevOps** for automated deployment to **Azure Container Registry** (ACR).
- Presented the MLOps proof-of-concept at the internal Data Engineering Marketplace, demonstrating production readiness and driving team adoption.

Summer Intern Jun 2022 – Jul 2022

- Designed an initial MLOps pipeline architecture for object detection and segmentation (Detectron2, YOLOv5) with DVC.
- Implemented ensemble strategies, performance visualizations, and automated evaluation reports to improve model robustness for autonomous driving scenarios.

Projects and Publications

[1] **Code Runtime Complexity Prediction -** ERCICA (Springer), 2023 | Python, TensorFlow, sklearn, Streamlit, NetworkX
Predicted Big-O runtime complexity of C/Java/Python code using static analysis and ML classification with BiLSTM over Abstract Syntax Trees graph embeddings on IBM CodeNet dataset (**Accuracy: 96%**). **[View Publication]**

[2] **Sentinel-Edge AI** - MadData Hackathon 2026 (Qualcomm Track) | Llama 3.2, Langchain, RAG, FastAPI
Selected as 1 of 10 teams to build an air-gapped **Edge AI** security auditor on Snapdragon X Elite NPU; engineered a RAG pipeline with ChromaDB to detect legal risks and code vulnerabilities without cloud egress. **[View Project]**

[3] **MLOps POC pipeline for Pedestrian Detection -** 2023 | Python, DVC, MLFlow, DAG, Streamlit, SQLite, PyTorch
Implemented on-premise MLOps starter template with DVC + Detectron2 for end-to-end ML lifecycle.**[View Project]**

[4] **Novel ways of decrypting transposition ciphers**, - IEEE smartgencon, 2022 | Python, Optimization, Natural Language
Introduced cipher-breaking optimization techniques over columnar transposition ciphers. **[View Publication]**

Teaching Experience

Teaching Assistant, CS564: Database Management Systems Madison, WI, USA
University of Wisconsin-Madison; Professor: Xiangyao Yu Jan 2026 – May 2026
Lead assignment sections, mentor students, and hold office hours to support database projects and assignments. (**SQL**, **Python**). Design and grade assignments, exams, and project rubrics; provide feedback to **200+** students to improve understanding of database design and query optimization. Assist course instructors with course logistics, and automated grading pipelines.

Teaching Assistant, CS203: Statistics for Data Science Bengaluru, India
PES University; Professors: Dr. Bharathi R, Swati M September 2022 – Dec 2022
Responsibilities included teaching, designing, and grading coursework, assignments, and projects. Delivered hands-on sessions on **Data Science topics** such as **hypothesis testing, statistics, and supervised ML**. Organized, managed, and served as a panelist for a Hackathon. Mentored **120+** students enrolled in the undergraduate level Data Science course.

Presentations and Competitions

MadData Hackathon 2026 (Qualcomm Track) Madison, WI, USA
University of Wisconsin-Madison & Qualcomm Feb 2026

- Selected as **1 of 10 teams** to build privacy-first Edge AI applications on the **Snapdragon X Elite NPU**.
- Developed "Sentinel-Edge AI," an air-gapped security auditor using local **Llama 3.2**, **Langchain** and **RAG** to analyze sensitive legal/code documents with zero cloud data egress.

3 Minute Thesis (3MT) Competition Madison, WI, USA
University of Wisconsin-Madison Nov 2025

- Selected to present research on "Generating Fine-Grained Datasets for Autonomous Driving" using vision foundation models.
- Condensed technical research into a compelling 3-minute narrative for a general, non-technical audience.

Professional Awards and Scholarships

Top Achiever (1st/150+) & Best Presenter Award	Bosch Data Engineering Dept.	2024
Distinction Award (Academic Excellence, Sem 3-6)	PES University	2022-2023
MRD Scholarship (40% Tuition, Top 20% of 1100+)	PES University	2020

Volunteering Experience

CSR Volunteer — Bengaluru, India
Bosch Global Software Technologies (BGSW) — Oct 2023 – Mar 2025

- **Tech Mentorship & Education:** Mentored candidates with disabilities in tech interview preparation (Youth4Jobs), taught Computer Science to underprivileged students, and developed interactive materials for rural childcare centers.
- **Social Impact:** Supported agricultural initiatives funding stipends for differently-abled individuals and conducted educational surveys at government boys' homes.
- **Community & Wildlife:** Assisted in elephant habitat restoration, volunteered for abandoned dog adoption drives (CUPA), and led infrastructure renovation and sports coaching at local government schools.